

An analysis of logic flaws in web-of-trust services

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Content

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Abstract

Web-of-trust services (WOT) such as Keybase, Onename, and Blockstack promise to verify individuals' identities on the web. Since many applications on the web are not consistent this often leads to unintended behaviour and therefore security vulnerabilities in web-of-trust services. This write-up analyses three attack vectors that I stumbled across while conducting research on the security of WOT services.

The Technology Behind WOT Services

WOT services allow users to create tokens and place them on their personal pages (e.g. GitHub profile). The service will then look for the token using a scraper and if the token is valid, display that the user does in fact own that external page. The idea behind this method is so that users can display what pages on the web belong to them and then tie their WOT account to all of those external services. On top of that, since the verification tokens need to be publicly accessible, other users can verify that the proof is legitimate by visiting the page containing the token.



tomnomnom

Tom Hudson

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UK

4 devices
CE03 0B9A DABB DC9E
5AD0 4EBF 0F16 2AAD
tomnomnom tweet
tomnomnom gist
tomnomnom post
tomnomnom profile
tomnomnom.com http
tomhudson.co.uk http
1HBk5H4yTxkyBmWPthxMYQdBwoiUEJ61cL

PGP Encrypt

Keybase Chat

User TomNomNom cryptographically verifying their online identity on Keybase.

Attack Vector One – Distribution Of Content On Timelines

On the web a lot of profile-based applications allow redistribution of content by sharing someone else's entry on your personal timeline – on Twitter users can *retweet* content and on GitHub people can *fork* projects. Since some WOT services use a public entry or post to verify the user's identity, I asked myself whether it would be possible to claim ownership of someone else's account by having them share a token that was posted onto an attacker's timeline. One particular service stood out for me, GitHub, where WOT applications such as Keybase require users to place the verification token into a GitHub gist file. The interesting behaviour that I noticed with GitHub is that when a user forks a gist, the gist is not only displayed on the user's timeline, it replaces the original author's username with the sharer's username.

bayotop / blockstack.md
Created 18 days ago — forked from EdOverflow/blockstack.md

1 file 0 forks 0 comments 0 stars

Verifying my Blockstack ID is secured with the address 1MYJhkNmKN6HFXUBpaeDfFja8j5bFR6VTr
<https://explorer.blockstack.org/address/1MYJhkNmKN6HFXUBpaeDfFja8j5bFR6VTr>

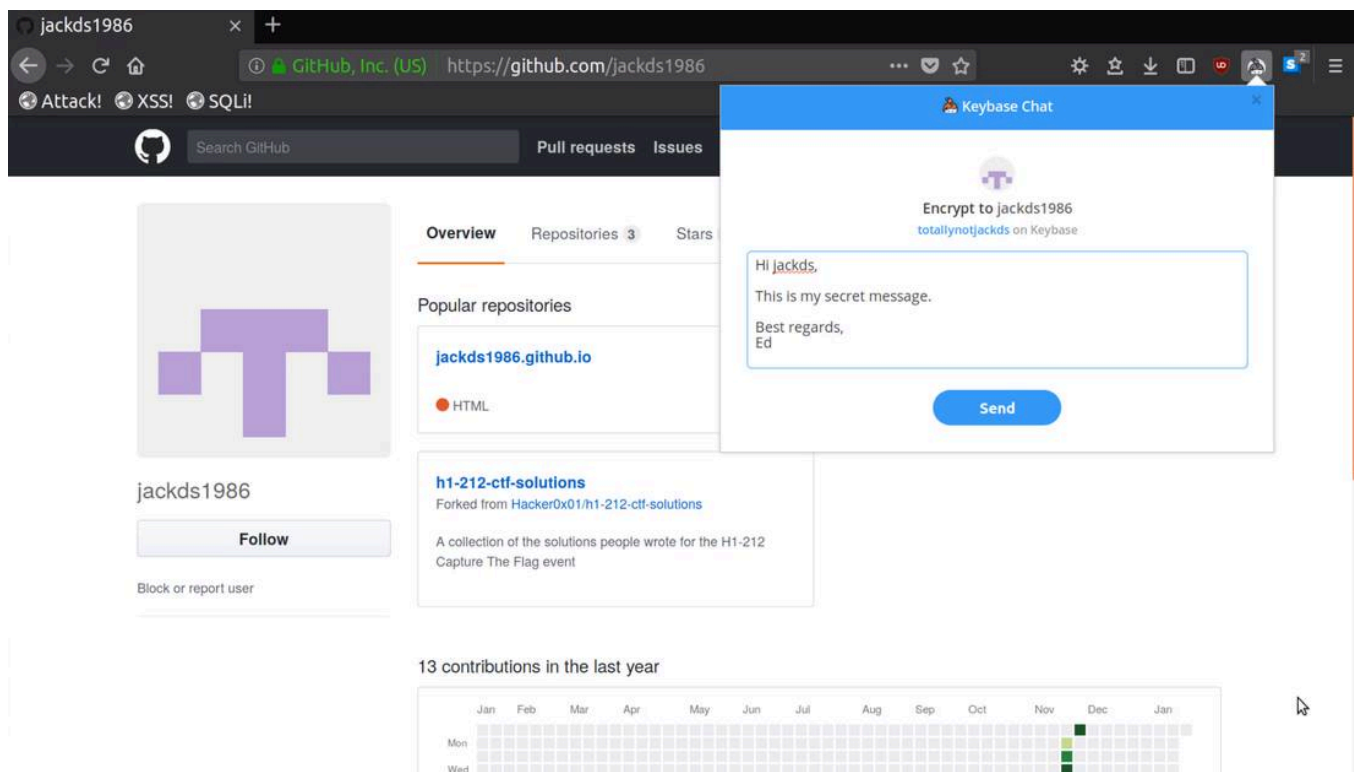
A forked GitHub gist — the original author is EdOverflow and the gist was forked by bayotop.

One vulnerable service was Keybase, where if an attacker could convince a victim to fork their GitHub gist, the attacker would be able to claim ownership of the victim's GitHub username. On top of that, Keybase allowed modification of the verification snippet, allowing an attacker to hide the token in an HTML comment.

An example attack using this technique against Keybase could have unfolded as follows:

- The attacker requests a verification token from Keybase for the victim's GitHub username;
- Keybase prompts the attacker to place the verification token in a `keybase.md` gist;
- The attacker creates a `keybase.md` gist hiding the verification token in an HTML comment;
- The victim forks the attacker's GitHub gist;
- The attacker instructs Keybase to verify `/<victim>/keybase.md`.

As a result, the attacker's Keybase account states that they own the victim's account. To make matters worse, Keybase has a [browser extension](#) that allows users to browse to certain applications (e.g. GitHub, Twitter, Hacker News, etc.) and message the user on Keybase via the profile page – the extension adds a little messaging window on the user's profile. Messages are sent to the account on Keybase that has verified ownership of the account. This means the extension will trick the user into thinking they are messaging the intended recipient, but all messages land in the attacker's inbox since they control the victim's username.



Hijacked GitHub username (@jackds1986) viewed in the Keybase browser extension. All messages are sent to a user called "totallynotjackds" on Keybase.

[Blockstack](#) and [fireblock.io](#) were also vulnerable to this attack vector. In most cases, the fix consisted of simply verifying whether the GitHub gist was a fork using GitHub's gist API.

Attack Vector Two – Namespace attacks

Some WoT services require placing the verification token in a specific filename. Keybase, for example, as mentioned in the previous section, require GitHub verification tokens to be placed in GitHub gists called `keybase.md`. On web assets, Keybase require the file to be named `keybase.txt` and either placed under the top-level directory or the `.well-known` path. The reason behind the

.well-known proposal in [RFC5785](#), is to prevent filename collisions and clogging up the root directory. The former is particularly interesting when it comes to WOT services since if an attacker can control the filename on a website, they could potentially claim ownership of the domain. One such case happened with liberapay.com and Keybase. Liberapay, an open-source donation platform, did not restrict username's containing dots in them; therefore one could create usernames containing file extensions. This became apparent to me when I set up a profile page for the security.txt project (<https://liberapay.com/security.txt>). I created a user called `keybase.txt` and embedded the Keybase verification snippet in the profile's description. This allowed me to claim ownership of liberapay.com. Keybase did not verify the content type of the keybase.txt file and did not even ensure that the token is not embedded into a page.

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keybase.txt

keybase.txt receives \$0.00 per week from 0 patrons.

[Donate](#)

Statement

```
-----BEGIN PGP MESSAGE-----
https://keybase.io/edoverflow
-----

I hereby claim:

* I am an admin of https://liberapay.com
* I am edoverflow (https://keybase.io/edoverflow) on keybase.
* I have a public key with fingerprint 0F42 60BF F680 87B2 9E3F 097B 7046 AE48 0053 3805

To do so, I am signing this object:

{
  "body": {
    "key": {
      "eldest_kid": "0128df1309f8c2e179c65c4a97bec3663bb5c5c59d294495dec14309d2b5eab5b74eda",
      "fingerprint": "0F426dbff68087b29e3fd97b7046ae480533805",
      "host": "keybase.io",
      "key_id": "7046ae480533805",
      "kid": "0101736376e1ae2fab309b6cc7dec1eeb493641b1c2e905a524f128b581b544bdeeb8a",
      "uid": "8860bf2e4a8602c8cf63f949ea3b319",
      "username": "edoverflow"
    },
    "service": {
      "hostname": "liberapay.com",
      "protocol": "https"
    },
    "type": "web_service_binding",
    "version": 1
  },
  "ctime": 1516307966,
  "expire_in": 157680000,
  "prev": "d4a7cc3ebaa5eac88e80506fa5ec70bd73e5a616ade651d9f291afa758245f7",
  "seegno": 17,
  "tag": "signature"
}

which yields the signature:

-----BEGIN PGP MESSAGE-----
Version: Keybase OpenPGP v2.0.76
Comment: https://keybase.io/crypto

yH8uAnIcBv3rUPVFLGQK1C0k4T1UE1BeYwE+7zPuUp128PogA00gzf1tvc5+BCV
yz3Xkey8ggyYFFIbpzf8NAFF9HC8A02bvDCM8umqBAG2AP8kstLfckdojYFoc+
a+8Fe9ba3/etb61236x42B7043yp7wV7tu7n09PVUGKyQLfCgrpaJL1rQCK8MD
75p2e6x78p0yU48ngRRLACapOCYFo1BAaK2RfHhRUEFLFeTVdMG2hJZF7FCs2x
g151x0b50QapFQ1pdkc8d1d8KcH6NNAxH8RZonSEASES8Jv103ppg4W20A0bn
WQnwHf1owu8QYegCyf2d31JgXX33v/gH/KctEgLnChGn3GAXuXHQFEQ2H4CND
1g4dRj8p5gY8S81Doxk2BY8GPNchF4w5Ltk5u65BAlp0oag2ABG88N8Vh5NqH
McTSgF8Y5cDFndCkqpe112aK5+J0HwK1+3KdgrdHLQ4zdrALOnfSkqIXE7rT
pXTBRBeTr8KPx+k2GIRFudP8AlnF8FahLRTa05K23QK5E26Y7K08Nk1rH255P0KZ
LLIqJF24r9PnwLabp8fFPhy1Dq4LELq8BQhcu1QhpDR21wnRAGN8hJ8QRITFVnH
Q4FV0UKFAk+rqQK8GpQ5AH8B2p1GX3tcehdkyQvYQ8Rdvs0HBT4s3U5ddecTH
PoGnx18J8R8BQasFJH5324N6m5hTX6+HXa6KLSLq1dJFvP6r/285pV88boe2+G
Hf9qVwOckmLf28q21H8Xy8N6ZXTWTFPnQQLD7v8kzrxtJ8Rq35r94N8h8P
nx+8czr73nd788qSkTn8N6a05903ZzH6N3NqAB3SutmenHvTfXr6Q2Awu8f2T7
/18dp46f89RdF86dX3d24cxf9fCc4hyQ1972na0Fad75T7R1z/R8XTpSL3U/XNk
9xGV7TK3rNLDpGuSB+Hbv8355FndC25+KvWwAK3zvP1DxrvPkes03LX+XPVA69
/UE0NAK3vr7H2XJx70109s6/SjWwKcNdvVg90y+V3H2h9v8ZneB3E+S+DEKxvN
8mq6v53dkxdt/88R8P32zCv+S8ndzXAZZr5to2uT2780xy20AdJpn8yZTFy7FR8J
750pA8PUNH8P8Gz+6sX3w+20d7K/jxqvH5zB37R1/e07Lp54yY82Co4T16fcty
rThqRjrdxLZLakVve0/un8y853K33KocLe8ArfX64Gxf87p8Hw7cvP+20KJ
wHw3y0d1tc38Pv8dy4ntL/RdhsxRFLd4e23bw98y01f/53Ttyu+89W84q8N
u/Penp7FL7Ww59ccp8084sC3T9T0SP25F17b36254462H4v2pc+H4TFTb4
2hu0tvG1017/2z5D8YxvT8=
=C/E2
-----END PGP MESSAGE-----
```

which yields the signature:

-----BEGIN PGP MESSAGE-----

Version: Keybase OpenPGP v2.0.76
Comment: https://keybase.io/crypto

```
yH8uAnIcBv3rUPVFLGQK1C0k4T1UE1BeYwE+7zPuUp128PogA00gzf1tvc5+BCV
yz3Xkey8ggyYFFIbpzf8NAFF9HC8A02bvDCM8umqBAG2AP8kstLfckdojYFoc+
a+8Fe9ba3/etb61236x42B7043yp7wV7tu7n09PVUGKyQLfCgrpaJL1rQCK8MD
75p2e6x78p0yU48ngRRLACapOCYFo1BAaK2RfHhRUEFLFeTVdMG2hJZF7FCs2x
g151x0b50QapFQ1pdkc8d1d8KcH6NNAxH8RZonSEASES8Jv103ppg4W20A0bn
WQnwHf1owu8QYegCyf2d31JgXX33v/gH/KctEgLnChGn3GAXuXHQFEQ2H4CND
1g4dRj8p5gY8S81Doxk2BY8GPNchF4w5Ltk5u65BAlp0oag2ABG88N8Vh5NqH
McTSgF8Y5cDFndCkqpe112aK5+J0HwK1+3KdgrdHLQ4zdrALOnfSkqIXE7rT
pXTBRBeTr8KPx+k2GIRFudP8AlnF8FahLRTa05K23QK5E26Y7K08Nk1rH255P0KZ
LLIqJF24r9PnwLabp8fFPhy1Dq4LELq8BQhcu1QhpDR21wnRAGN8hJ8QRITFVnH
Q4FV0UKFAk+rqQK8GpQ5AH8B2p1GX3tcehdkyQvYQ8Rdvs0HBT4s3U5ddecTH
PoGnx18J8R8BQasFJH5324N6m5hTX6+HXa6KLSLq1dJFvP6r/285pV88boe2+G
Hf9qVwOckmLf28q21H8Xy8N6ZXTWTFPnQQLD7v8kzrxtJ8Rq35r94N8h8P
nx+8czr73nd788qSkTn8N6a05903ZzH6N3NqAB3SutmenHvTfXr6Q2Awu8f2T7
/18dp46f89RdF86dX3d24cxf9fCc4hyQ1972na0Fad75T7R1z/R8XTpSL3U/XNk
9xGV7TK3rNLDpGuSB+Hbv8355FndC25+KvWwAK3zvP1DxrvPkes03LX+XPVA69
/UE0NAK3vr7H2XJx70109s6/SjWwKcNdvVg90y+V3H2h9v8ZneB3E+S+DEKxvN
8mq6v53dkxdt/88R8P32zCv+S8ndzXAZZr5to2uT2780xy20AdJpn8yZTFy7FR8J
750pA8PUNH8P8Gz+6sX3w+20d7K/jxqvH5zB37R1/e07Lp54yY82Co4T16fcty
rThqRjrdxLZLakVve0/un8y853K33KocLe8ArfX64Gxf87p8Hw7cvP+20KJ
wHw3y0d1tc38Pv8dy4ntL/RdhsxRFLd4e23bw98y01f/53Ttyu+89W84q8N
u/Penp7FL7Ww59ccp8084sC3T9T0SP25F17b36254462H4v2pc+H4TFTb4
2hu0tvG1017/2z5D8YxvT8=
=C/E2
-----END PGP MESSAGE-----
```

And finally, I am proving ownership of this host by posting or
appending to this document.

View my publicly-auditable identity here: <https://keybase.io/edoverflow>

History

keybase.txt joined 2 days ago.

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Claiming ownership of liberapay.com by creating a user called keybase.txt and embedding the verification snippet into the profile's description.

Attack Vector Three – Redirects

After claiming ownership of liberapay.com I noticed that liberapay.org redirects to liberapay.com. This next attack consisted of using a verification token generated for liberapay.org embedded on liberapay.com to claim ownership of liberapay.org. Keybase's scraper would blindly follow the redirect and not validate the final endpoint to make sure it matches the target host. Keybase would request liberapay.org/keybase.txt which redirects to liberapay.com/keybase.txt where a valid keybase.txt file is located.



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::1 

 2 devices
 7046 AE48 0053 3805  edit
 edoverflow  tweet
 prove your Facebook identity
 bayotop  gist
 prove your Reddit identity
 prove your Hacker News identity
 edoverflow.com  https!
 liberapay.com  https
 liberapay.org  https
 prove another web identity
 1Bcescpwo46PKVFBgQBL7BZdLCZhCDs9Nv
 set a Zcash address

[PGP Encrypt](#) [Keybase Chat](#)

Claiming ownership of liberapay.org via liberapay.com's keybase.txt file.

One particular plausible attack scenario that I could come up with was claiming branded URL-shorteners using this technique.

Conclusion

All services affected by these attack vectors were notified and promptly resolved most of the reported issues. Keybase remain vulnerable to attack vectors 2 and 3 – as far as I can tell they do not plan on resolving those issues. I was thoroughly impressed by the response times of all the affected parties and I look forward to working with them again in the future. As a result of this research, I have become addicted to finding logic flaws.

Update (Friday, 16 February 2018): [@LewisBugBounty](#) demonstrated that one can claim ownership of URL shorteners as I theorised above: [Tweet](#).



Lewis

@LewisBugBounty

Follow



Keybases lack of checks for Site verification is worrying

Find a URL Shortener site which allows for 'keybase.txt' as a custom URL and then have it redirect to a /raw/ pastebin of your verification to claim yourself a nice site.

@EdOverflow Showed something like this awhile ago

